

A **token** is the smallest individual unit in a C program. The C compiler breaks source code into tokens during lexical analysis. There are 6 types of tokens:

Token Type	Description	Examples
Keywords	Reserved words with fixed meaning	int, float, if, for, return, void, struct
Identifiers	Names given to variables, functions, etc.	main, total, printf, myVar
Constants	Fixed values that don't change	42, 3.14, 'A', "hello"
String Literals	Sequence of characters in quotes	"Hello World", "C Programming"
Operators	Symbols performing operations	+, -, *, /, %, ==, &&,
Punctuators	Special symbols used structurally	{ } () [] ; , . ->

■ Keywords (32 in C89/ANSI)

auto	break	case	char	const	continue	default	do
double	else	enum	extern	float	for	goto	if
int	long	register	return	short	signed	sizeof	static
struct	switch	typedef	union	unsigned	void	volatile	while

■ *Note: Keywords cannot be used as variable names. C is case-sensitive — 'Int' is not a keyword, but 'int' is.*

■ Identifiers – Naming Rules

- Must begin with a letter (a–z, A–Z) or underscore _
- Can contain letters, digits (0–9), and underscores
- Cannot be a keyword
- Case-sensitive: total ≠ Total ≠ TOTAL
- No spaces or special characters (!@#\$%^&* etc.)
- No length limit (significant up to 31 characters in C89)

```
// Valid identifiers
int age;
float _price;
char firstName;
double totalAmount2;

// Invalid identifiers
int 2ndValue;    // Cannot start with digit
float my-var;    // Hyphens not allowed
char int;        // 'int' is a keyword
```

Data Type	Size (bytes)	Range	Format Specifier
char	1	-128 to 127	%c
unsigned char	1	0 to 255	%c
short	2	-32,768 to 32,767	%hd
unsigned short	2	0 to 65,535	%hu
int	4	-2,147,483,648 to 2,147,483,647	%d
unsigned int	4	0 to 4,294,967,295	%u
long	4/8	Platform-dependent	%ld
long long	8	-9.2×10 ¹⁸ to 9.2×10 ¹⁸	%lld
float	4	±3.4×10 ³⁸ (6 decimals)	%f
double	8	±1.7×10 ³⁰⁸ (15 decimals)	%lf
long double	10/16	Extended precision	%Lf
void	0	No value	—

```
#include <stdio.h>
#include <limits.h>    // INT_MAX, CHAR_MAX etc.
#include <float.h>     // FLT_MAX, DBL_MAX etc.

int main() {
    // Variable declaration and initialisation
    int    age      = 21;
    float  pi        = 3.14159f;
    double precise = 3.141592653589793;
    char   grade     = 'A';
    char   name[]    = "Alice";

    // sizeof operator
    printf("Size of int      : %zu bytes\n", sizeof(int));
    printf("Size of double : %zu bytes\n", sizeof(double));
    printf("INT_MAX = %d\n", INT_MAX);
    return 0;
}
```

Type Qualifiers

Qualifier	Meaning	Example
const	Value cannot change after init	const float PI = 3.14159f;
volatile	Value may change externally (hardware)	volatile int sensor;

Qualifier	Meaning	Example
restrict	Pointer is sole access path (C99)	void f(int *restrict p);
signed	Explicitly signed (default for int/char)	signed int temp = -5;
unsigned	Only non-negative values	unsigned int count = 0;